Training Feedback Tailored to Technical Skills Expedites Proficiency in Performing Watertight Vesicourethral Anastomoses: A Randomized Controlled Trial

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BACKGROUND AND OBJECTIVE: The vesicourethral anastomosis (VUA) is a technically demanding step during the robot-assisted radical prostatectomy (RARP), requiring a watertight seal to prevent postoperative urine leak. Our randomized controlled trial evaluated suturing skills critical to successful VUAs and assessed whether tailored feedback focusing on deficient skills expedites their proper execution.

METHODS: Nineteen residents completed weekly simulated robot-assisted VUAs on silicone bladder-urethra models for 10 weeks, with a follow-up session after 3 to 4 weeks. All received standardized basic training and completed baseline sessions. We evaluated technical performance using our previously validated End-to-End Assessment of Suturing Expertise (EASE) tool. Participants were randomized into 3 feedback groups: (1) *Control* (training video), (2) *Conventional* (training video + EASE scoring criteria + video recordings of their own previous sessions), and (3) *Tailored* (all *Conventional* resources + feedback on key deficient skills).

KEY FINDINGS AND LIMITATIONS: We had 17 of 19 participants complete all 10 sessions ($n = 170$). All 7 participants in the *Tailored* group achieved and sustained watertight VUAs, compared to only 1 of 6 in the *Conventional* and 0 of 5 in the *Control* groups ($p < 0.001$). Suturing skill metrics predicted watertightness with 78% sensitivity (95% confidence interval 0.70-0.86) and 72% specificity (95% confidence interval 0.61-0.82). The top 5 suturing skills predicting a watertight VUA included *Needle Hold Angle*, *Depth of Needle Hold*, *Needle Repositions*, *Wrist Rotation during Needle Withdrawal*, and *Needle Driving Smoothness*. The *Tailored* group retained their performance at the follow-up session, 3 to 4 weeks later.

CONCLUSIONS AND CLINICAL IMPLICATIONS: Tailored feedback targeting deficient skills significantly expedites achieving VUA watertightness, and surgical skills quantified by the EASE tool can predict a successful VUA. (J Surg Ed 83:103937. © 2026 Association of Program Directors in Surgery. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.)

KEY WORDS: Feedback, Simulation, Urology, Robotic, Surgery, Resident

ABBREVIATIONS: VUA, vesicourethral anastomosis; RARP, robot-assisted radical prostatectomy; EASE,

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End-to-End Assessment of Suturing Expertise; IRR, Interrater reliability; PABAK, prevalence-adjusted bias-adjusted kappa; GEE, generalized estimating equations

INTRODUCTION

The vesicourethral anastomosis (VUA) is a critical reconstructive step during the robot-assisted radical prostatectomy (RARP), where failure to achieve watertightness between the bladder and the urethra can result in urine leak with significant sequelae. Complications include uroperitoneum, ileus, and bladder neck contractures. The postoperative leak rate at 3 weeks ranges from 8% to 14.1% (Tyritzis,[1]).

Growing evidence demonstrates that surgical performance relates directly to patient outcomes (Birkmeyer,[2]; Hung,[3]). To evaluate surgeon performance for suturing, our group previously developed and validated the End-to-End Assessment of Suturing Expertise (EASE) tool (Supplemental figure 1) to identify and objectively evaluate key technical skills (Haque,[4]). When applying our assessment tool to prior recordings of live surgical procedures, we demonstrated EASE's clinical value through a significant association between our defined technical skills and the procedures' resulting clinical outcomes, such as urinary continence recovery after RARP (Chu,[5]).

Using performance metrics to assess the RARP, an efficient dry laboratory model exercise can be used to teach and progress trainees into mastering the surgery (Ticonosco,[6]). In a previous dry lab trial, we showed that it is feasible to deliver focused feedback to surgical trainees on only those skills where they are deficient, and this *Tailored* feedback may expedite the acquisition of robotic suturing skills when compared with trainees receiving standard training without feedback (Ma[7]). This work, however, evaluated suturing skills and not their application specifically to the VUA in a RARP. During training, improving and mastering technical skills is vital to minimizing these adverse outcomes, more so for surgeries with higher complication rates (Stulberg,[8]).

In the present study, we conducted a randomized controlled trial to identify and evaluate the critical technical skills for determining a watertight VUA and further investigated whether feedback tailored to specific deficient skills during training can expedite the achievement of successful watertightness.

MATERIALS AND METHODS

Study Design Overview

In order to see how different modes of training feedback influence skill acquisition, we recruited 19 surgical residents from Urology and General Surgery to perform robot-assisted suturing exercises, connecting standardized silicone bladder and urethra models (3-Dmed) simulating VUA post-RARP. Eight participants were naïve to robotic suturing and had never performed suturing at the console. Only 2 urology PGY-5 residents (one per feedback arm) had substantial prior experience performing the VUA during RARPs, while 2 PGY-4 residents had minimal exposure, limited to 1 to 2 VUA attempts. All remaining participants had no prior VUA experience. Participants completed 10 sessions over 12 weeks, with the first serving as a baseline for randomization. A retention assessment was performed 3 to 4 weeks after completion of the curriculum, consistent with prior studies evaluating short-term retention of robotic and simulation-based surgical skills (Jenison,[9]; Vamadevan,[10]) (Fig. 1). All trainees were given a prestudy survey where baseline demographics were collected, as shown in Table 1.

Study Protocol in Detail

Participants received standardized training for the da Vinci XI surgical robot, then completed a baseline suturing session after viewing an ideal video demonstration of an attending surgeon performing the VUA exercise. The exercise consisted of 16 stitches, 8 on the silicone bladder and 8 on the silicone urethra. Each session was rated using the EASE suturing skill assessment tool, and models were evaluated for watertightness. After the baseline suturing session, principal component analysis extracted independent skill components, followed by cluster analysis to group subjects into 2 clusters based on baseline skill. Stratified randomization then assigned subjects into 3 groups based on baseline performance: (1) *Control* (access to the ideal performance training video without feedback provided after each session), (2) *Conventional* feedback (ideal training video, a reference for all EASE scoring criteria, and video recordings of each of their sessions for review), and (3) *Tailored* feedback (all other resources plus their assessment results for deficient technical skills) (Fig. 2). The *Tailored* feedback group received scores for selected skills rated as nonideal and referenced against the ideal EASE criteria, which was provided to both the *Conventional* and *Tailored* groups. All participants, regardless of group, completed a total of 10 weekly VUA exercises.

To approximate clinical relevance, we assessed watertightness after each VUA exercise by instilling 20 mL of water into the completed model and measuring leakage. Watertightness was defined as less than 2 mL of liquid leaked after 20 seconds. We assessed the sustainability of watertightness over multiple sessions, defined by achieving at least 2 consecutive sessions of

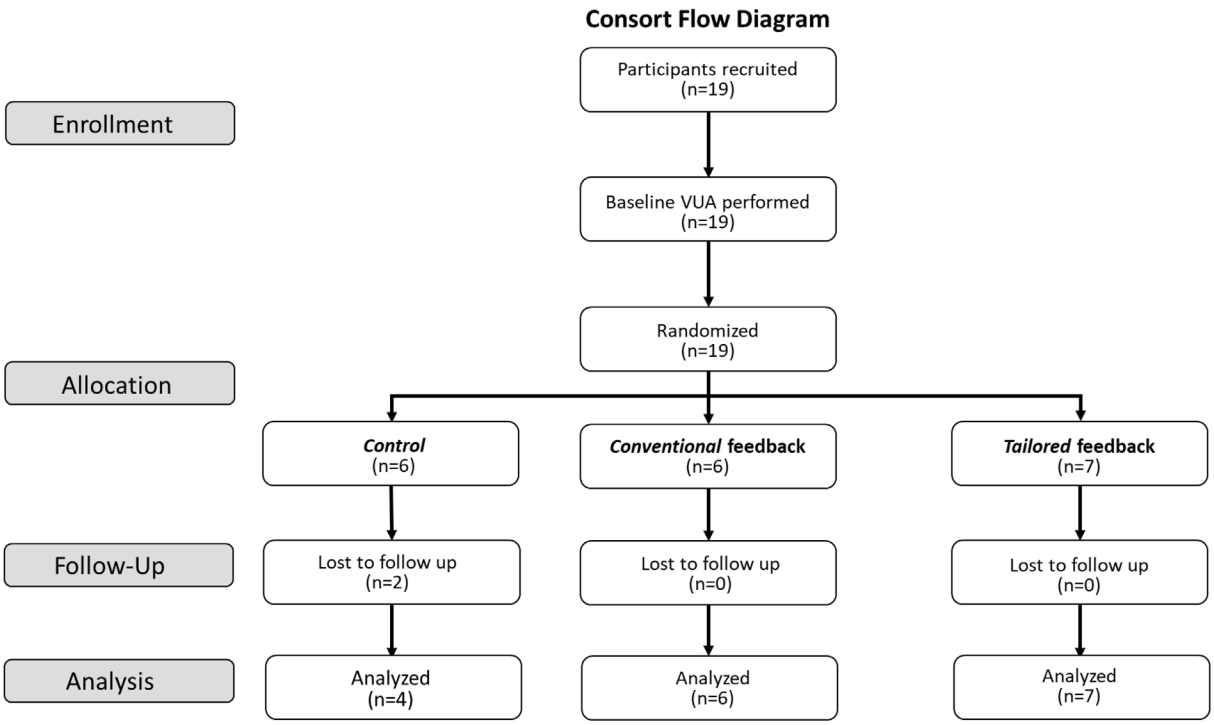


FIGURE 1. Consort flow diagram showing participants progressing through the randomized controlled trial.

watertightness and maintaining these outcomes through the remainder of the study.

At the study conclusion, all participants returned after 3 to 4 weeks to perform one additional simulation session to assess retention of suturing skills. All groups completed this retention session without further instruction. Performance was gauged via technical skills assessment and watertightness testing as initially evaluated.

Inter-rater Reliability During Technical Skills Evaluation

Suturing performance was evaluated using the EASE tool for technical skills assessment. Two independent blinded raters received standardized training for EASE and performed these technical skills assessments for each recorded exercise ($n = 170$). Inter-rater reliability was calculated using prevalence-adjusted bias-adjusted kappa. For each skill evaluated, the prevalence-adjusted bias-adjusted kappa ranged from 0.8, 95% confidence interval (CI 0.78, 0.82) to 0.99, 95% CI (0.99, 1), and each skill was only rated as "ideal" performance when both raters concurred. EASE scores were compared across the 3 feedback groups based on the proportion of ideal technical skill scores during the suturing exercise.

Statistical Analysis

Randomization for feedback groups was performed after baseline assessment. To balance technical skills across groups, we used exploratory factor analysis with varimax rotation to detect latent factors under EASE scores. K-means analysis then identified clusters based on latent factors. A stratified randomization approach balanced cluster distribution representing skill levels across the 3 groups. The exploratory factor analysis was performed to support stratified randomization across skill domains. The specific factor loadings and structure are not reported because no latent inference was intended.

Our primary outcome was watertightness, defined as <2 mL leakage after 20 seconds of instilling 20 mL water into the sutured silicone model. We then compared watertightness between feedback groups. Hierarchical Poisson regression models with robust generalized estimating equations were used to fit the model by considering the nested data structure of participant and session and the right skewness of net water retention. Given that watertightness occurred frequently in our study, we selected a modified Poisson regression approach to directly estimate rate ratios between intervention groups rather than an odds ratio. We assumed an exchangeable covariance structure. Quasi-Information Criterion compared the model fit across different covariance structures.

For the performance outcome, we conducted 9 contrast tests to compare the change from pretraining (baseline) to post-training (session 10), pretraining to follow-up (retention session 11), and post-training to follow-up

TABLE 1. Baseline Difference by Group

Demographics	Total	*Control*	*Conventional* feedback	*Tailored* feedback
Number	*n* = 19	*n* = 6	*n* = 6	*n* = 7
Median yrs age (IQR)	30 (28-32)	29 (27-33)	29 (28-33)	30 (29-31)
No. gender (%)				
Male	10/19 (53%)	3/6 (50%)	2/6 (33%)	5/7 (71%)
Female	9/19 (47%)	3/6 (50%)	4/6 (67%)	2/7 (29%)
Median prior caseload (IQR)	4 (0-10)	0 (0-8)	5 (0-15)	5 (2-10)
Median yrs surgical training (IQR)	2 (1-3)	1 (1-2)	1.25 (1-3)	3 (2-4)
No. training level (%)				
PGY1	5/19 (26%)	3/6 (50%)	2/6 (33%)	0/7 (0%)
PGY2	5/19 (26%)	1/6 (17%)	2/6 (33%)	2/7 (29%)
PGY3	3/19 (16%)	1/6 (17%)	0/6 (0%)	2/7 (29%)
PGY4	4/19 (21%)	1/6 (17%)	1/6 (17%)	2/7 (29%)
PGY5	2/19 (11%)	0/6 (0%)	1/6 (17%)	1/7 (14%)
Rate of ideal Suturing sub-skill (95% IC)				
Needle repositions	60.3% (50.5%, 70.1%)	57.7% (34.7%, 80.7%)	70.8% (58.4%, 83.3%)	53.6% (42.4%, 64.8%)
Needle hold ratio	70.1% (59.7%, 80.6%)	57.4% (30.7%, 84.2%)	75.0% (65.9%, 84.1%)	76.8% (66.6%, 87.1%)
Needle hold angle	74.2% (63.3%, 85.1%)	69.0% (42.2%, 95.8%)	78.1% (65.6%, 90.6%)	75.4% (60.8%, 89.9%)
Depth of needle hold	73.4% (63.6%, 83.3%)	55.6% (34.9%, 76.4%)	88.5% (83.2%, 93.9%)	75.7% (64.9%, 86.6%)
Needle driving smoothness	74.6% (62.2%, 87.0%)	54.4% (23.3%, 85.6%)	86.5% (76.3%, 96.6%)	81.6% (73.8%, 90.0%)
Wrist rotation during needle driving	72.8% (62.5%, 83.1%)	58.6% (32.8%, 84.5%)	85.4% (78.5%, 92.3%)	74.3% (64.9%, 83.6%)
Wrist rotation during needle withdrawal	51.1% (39.8%, 62.3%)	40.1% (16.5%, 63.6%)	54.2% (35.5%, 72.8%)	57.8% (44.4%, 71.3%)
Needle targeting	81.7% (72.0%, 91.5%)	69.0% (43.9%, 94.0%)	92.7% (86.6%, 98.8%)	83.3% (74.5%, 92.1%)
Suture awareness	86.0% (76.1%, 95.8%)	76.3% (47.9%, 100%)	94.8% (90.3%, 99.2%)	86.6% (81.0%, 92.3%)
Cinching technique	55.6% (46.4%, 64.8%)	48.4% (29.8%, 66.9%)	65.6% (51.0%, 80.3%)	53.2% (41.4%, 65.0%)

for *Tailored* vs *Control*, *Tailored* vs *Conventional*, and *Conventional* vs *Control*. Bonferroni adjustment corrected the type I error. An intention-to-treat protocol was followed.

For EASE technical skills measurements, we compared the count of ideal skills observed for each of the 8 suturing tasks using the same approach with Hierarchical Poisson regression. A Classification Tree with 10-fold cross-validation was used to build a prediction model between EASE skills and watertightness. The variable of importance score was determined by the total reduction in impurity via the Gini index. Salford Predictive Modeler was used for CART analysis, and SAS9.4 for all other analyses.

We used an identity link function rather than the traditional log link to estimate Absolute Rate Differences. Absolute Rate Differences is clinically interpretable and more meaningful for medical decision-making than rate ratios from a log link function. Because the identity link does not constrain predicted values or CIs to non-negative bounds, negative estimates were truncated at zero.

RESULTS

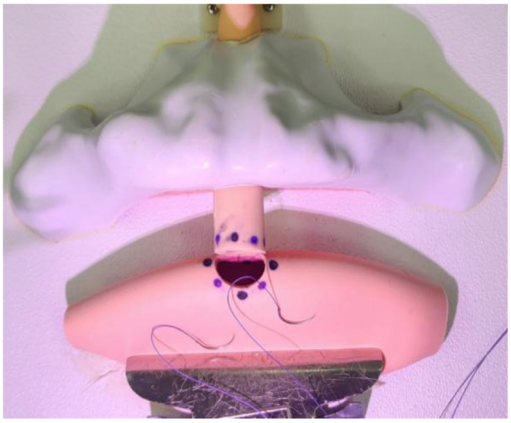
Demographic Data of Participants

Nineteen residents were recruited for the study. Table 1 describes their demographic characteristics. Two participants in the *Control* group dropped during the first 4 sessions due to scheduling conflicts, leaving 17 of 19 participants completing all 10 weekly suturing sessions ($n = 170$ total sessions) and one additional retention session ($n = 17$). The conventional feedback group demonstrated notably higher skill levels, the control group showed the lowest, and the tailored feedback group was intermediate.

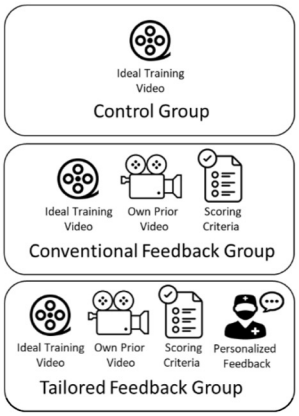
Achieving Watertightness

A statistically significant interaction for watertightness was seen across the 3 groups and 3 timepoints (pretraining, post-training, follow-up) ($p = 0.002$). The *Tailored* feedback group showed greater improvement from session 1 (pretraining) to session 10 (post-training) when

A)



B)



C)

End-to-end **A**ssessment of **S**uturing **E**xpertise
Phases, Skills, and Skill Grading Criteria

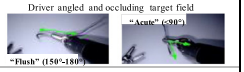
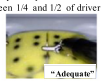
Phase	Skill	Ideal (3)	Adequate (2)	Fail (1)
Needle Handling	Needle Repositions	2 or less re-positions of needle prior to driving (discrete changes in needle position involving both drivers) (i.e. passing from hand to hand)	3 re-positions of needle prior to driving (discrete changes in needle position involving both drivers) (i.e. passing from hand to hand)	4 or more re-positions of needle prior to driving (discrete changes in needle position involving both drivers) (i.e. passing from hand to hand)
	Needle Hold Ratio	Surgeon grasps needle at or inside of acceptable range (from tip, between halfway and 3/4 length of needle)	---------	Surgeon grasps needle outside of acceptable range (from tip, within first half of needle or beyond 3/4 length)
	Needle Hold Angle	Needle tip to driver angle slightly obtuse (90° - 135°)	Needle tip to driver angle obviously obtuse (135° - 150°)	Needle tip to driver angle flush (150° - 180°) or acute (< 90°) **OR** Driver angled and occluding target field 
	Depth of Needle Hold	Needle held at tip of driver (distal 1/4 of driver jaws)	Needle held beyond tip but before middle of driver (between 1/4 and 1/2 of driver jaws) 	Needle held beyond middle of driver (at or beyond 1/2 length of driver jaws)
Needle Entry	Needle Entry Angle	Needle tip angle perpendicular (45° - 90°) to tissue plane	---------	Needle tip angle greater than 90° or less than 45° relative to tissue plane
Needle Driving	Driving Smoothness	Smooth, continuous motions **AND** Maximum 1 adjustment of needle trajectory / exit point with some possible reversal **AND** Maximum 1 release and re-grab of needle **AND** No complete removal of needle from tissue	2-3 adjustments of needle trajectory / exit point with some possible reversal **OR** 2 or more re-grabs of needle **AND** No complete removal of needle from tissue	4 or more adjustments of needle trajectory / exit point **OR** Complete removal of needle from tissue
	Wrist Rotation	Full wrist rotation while moving through tissue **AND** No lateral deviation of needle **AND** Trajectory follows curve of needle **AND** No tissue trauma	Partial wrist rotation while moving through tissue **OR** Some lateral deviation of needle **OR** Needle drive trajectory does not follow curve of needle **OR** Some likely tissue trauma but no tissue tear	No wrist rotation **OR** Definite tearing of tissue
Needle Withdrawal	Wrist Rotation	Full wrist rotation while moving through tissue **AND** No lateral deviation of needle **AND** Trajectory follows curve of needle **AND** No tissue trauma	Partial wrist rotation while moving through tissue **OR** Some lateral deviation of needle **OR** Needle drive trajectory does not follow curve of needle **OR** Some likely tissue trauma but no tissue tear	No wrist rotation **OR** Definite tearing of tissue
Suture Placement and Management	Suture Awareness	Suture does not get caught on driver / needle instrument **AND** Suture does not loop or tangle on itself unintentionally **AND** Suture is not "locked" or tied between suture throws	---------	Suture gets caught on driver / needle instrument **OR** Suture loops or tangles on itself unintentionally **OR** Suture becomes "locked" or tied between stitches **OR** Suture breaks
	Cinching Technique	Suture drawn through tissue smoothly **AND** Suture re-grasped incrementally and near suture site **AND** Suture tightened by pulling perpendicular to suture site **AND** No loose suture between suture throws	---------	Suture drawn through tissue roughly **OR** Suture tightened in very long pulls, possibly causing instruments out of view **OR** Suture grasped and cinched far from suture site **OR** Suture left loose between suture throws

FIGURE 2. VUA model, feedback groups, and EASE Rubric. (A) Silicone urethra and bladder model on which participants performed the robotic suturing exercise. (B) Descriptions of the 3 feedback groups into which participants were randomized: (1) *Control* (ideal training video), (2) *Conventional* (training video, EASE scoring criteria, video recordings of their previous sessions available for review), and (3) *Tailored* (all resources available to the *Conventional* group plus assessment score results matched to key suturing skills). *Tailored* feedback involved showing the participant examples of deficient technical suturing skills from their prior recorded session that were scored as nonideal, and referencing the criteria that would qualify that skill as ideal. C) EASE scoring rubric that was available for those in the *Tailored* feedback group to review.

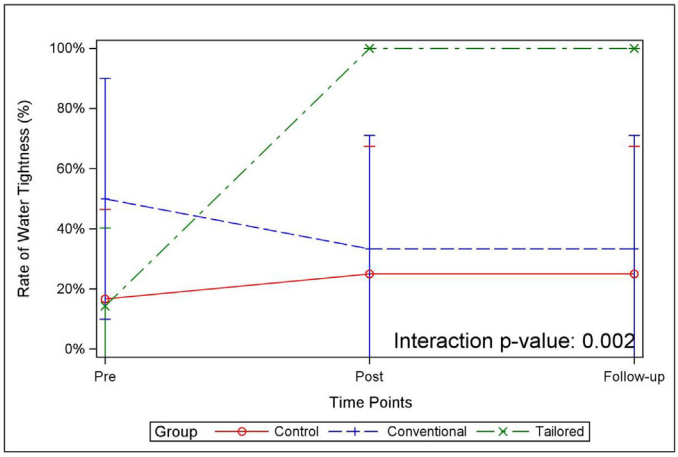


FIGURE 3. Group differences at 3 timepoints. Interaction test of watertightness comparing the different feedback groups at the time points of pretraining, post-training, and follow-up.

compared with the *Conventional* and *Control* groups (Fig. 3). The difference in percent improvement from pre- to post-training between the *Tailored* and *Control* groups was 60.71% (10.99%, 110.44%) ($p = 0.02$). However, after Bonferroni adjustment for 9 sets of contrast tests, the p value reduced to 0.15. The difference in the pretraining to post-training improvement between the *Tailored* and *Conventional* groups was 102.38% (41.59%, 163.17%) $p < 0.001$, where the performance for the *Tailored* group largely improved, while the *Conventional* group worsened in outcome (Supplemental Table 1, Fig. 3). The adjusted p value remained statistically significant ($p = 0.009$).

Similar trends were observed for net fluid retained, though not statistically significant (Supplemental Table 1).

To understand the trajectory of suturing skills across sessions, we illustrated the percent watertightness by the 3 feedback groups (Fig. 4). All 3 groups had very different trajectory patterns via an interaction test ($p < 0.001$). All *Tailored feedback* participants reached the goal of watertightness by session 8; in contrast, neither the *Conventional* feedback nor *Control* group was close to this level of performance, where only 1 participant achieved watertightness in the *Conventional* feedback group, and no participants in the *Control* group achieved watertightness.

To understand the mechanism behind performance improvement in watertightness, we used a Classification Tree to identify the suturing skills that influence watertightness through a 10-fold cross-validation. Our model showed a sensitivity of 0.78, 95% CI (0.7, 0.86), and

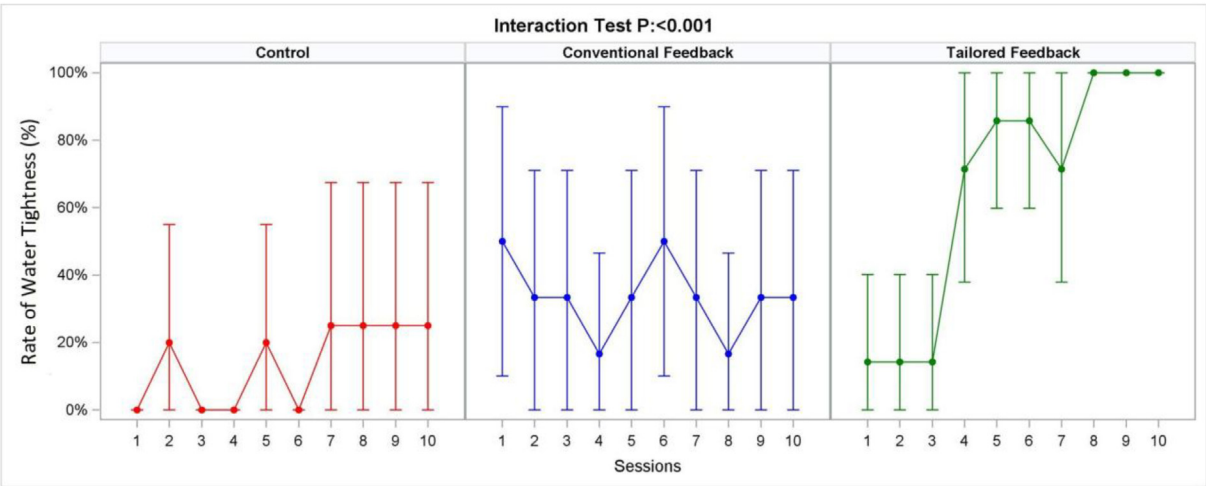


FIGURE 4. Trajectory of watertightness through 10 training sessions. Each dot represents the median watertightness outcome among participants for the representative session, with the error bars as the interquartile range.

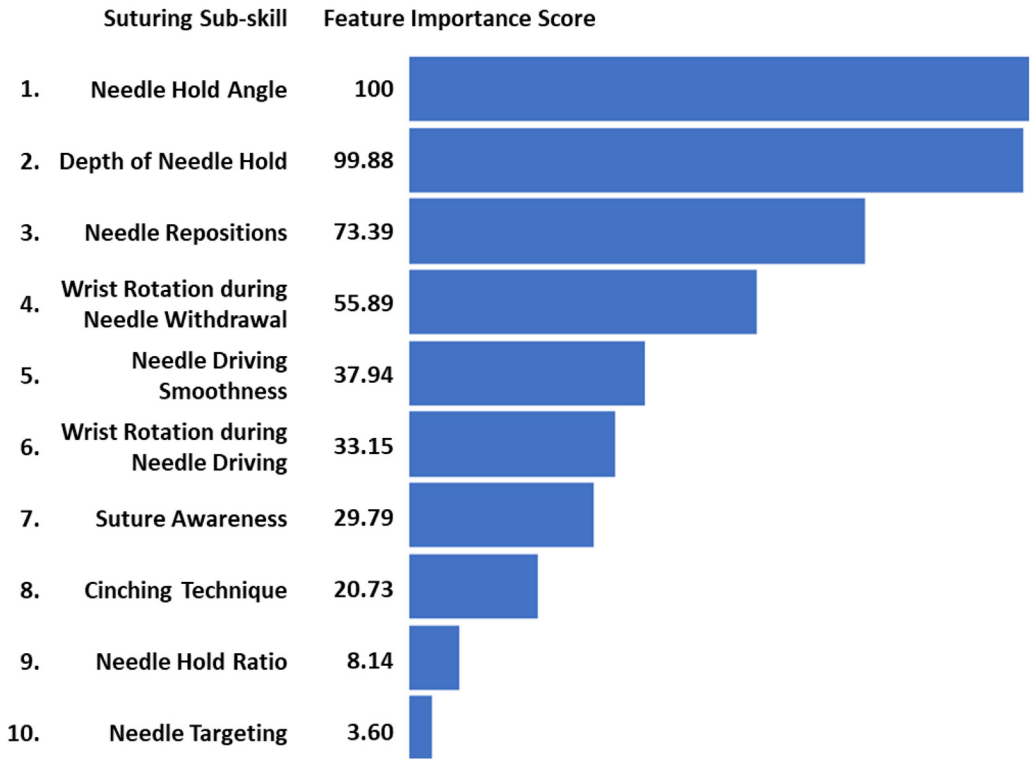


FIGURE 5. Does EASE skill scores predict outcome? Variables of importance scored and ranked by the gain or loss of function in predicting watertightness.

specificity of 0.72, 95% CI (0.61, 0.82) in predicting watertightness. Notably, the top 5 EASE suturing technical skills that helped predict a watertight VUA included: *Needle Hold Angle*, *Depth of Needle Hold, Needle Repositions*, *Wrist Rotation during Needle Withdrawal*, and Needle *Driving Smoothness* (Fig. 5).

We further compared improvement trajectory patterns of individual EASE skills. *Depth of Needle Hold*, *Needle Repositions*, and *Cinching Technique* showed the most rapid improvement ($p < 0.05$ interaction test with df = 20, Fig. 6, Supplementary Table S3). *Wrist Rotation during Needle Withdrawal, Needle Driving Smoothness,* and *Needle Hold Angle* did not show a statistically significant interaction for the global difference, but the *Tailored* feedback group achieved higher rates of ideal performance by session 10, not seen in the other groups (Fig. 6). All 6 of these skills reached higher ideal rates by session 10 for the *Tailored* group compared to the other 2 groups. *Needle Hold Angle* and *Needle Repositions*, for example, reached 99.1% (95% IC: 97.48%, 100%) and 98.21% (95% IC: 96.12%, 100%) ideal rates, respectively, for the *Tailored* group, while other groups did not reach similar levels. The difference within the *Conventional* group for *Needle Hold Angle* and *Needle Repositions* at session 10 remained statistically significant after Bonferroni adjustment for 30 pairwise comparisons ($p < 0.001$). In terms of improvement rate, the *Tailored* group reached higher rates of ideal scores earlier than the 2 other groups. For example, *Needle Hold Angle* and *Needle Repositions* reached 96.42% (95% IC: 94.13%, 98.72%) and 91.96% (95% IC: 86.05%, 97.88%) ideal scores, respectively in the *Tailored* group rapidly as early as sessions 2 and 3, while the *Conventional* and *Control* groups achieved similar rates much later (or not at all) by session 9. The estimated ideal performance rate for each EASE skill at each session, plus pairwise group differences, is presented in Supplementary Table 2.

Importantly, all 6 EASE skills that had consistently higher ideal rates for the *Tailored* feedback group were also top predictors for watertightness (Figs. 5 and 6).

After 10 weekly sessions were completed, participants performed one final suturing session to assess retention of acquired skills 3-4 weeks later. All 7 of 7 (100% [100%, 100%]) *Tailored* feedback participants again achieved watertightness, compared to 2 of 6 (33.33% [−4.39%, 71.05%]) *Conventional* group and only 1/4 (25% [−17.43%, 67.44%]) *Control* group participants. There was no change in the rate of watertightness found when comparing the 10th session to the retention follow-up for all 3 groups, meaning all *Tailored* feedback participants-maintained watertightness, while other groups did not improve or declined (Fig. 1, Supplementary Table 1). This outcome was affirmed by retention in EASE domains (Fig. 6). There were no performance

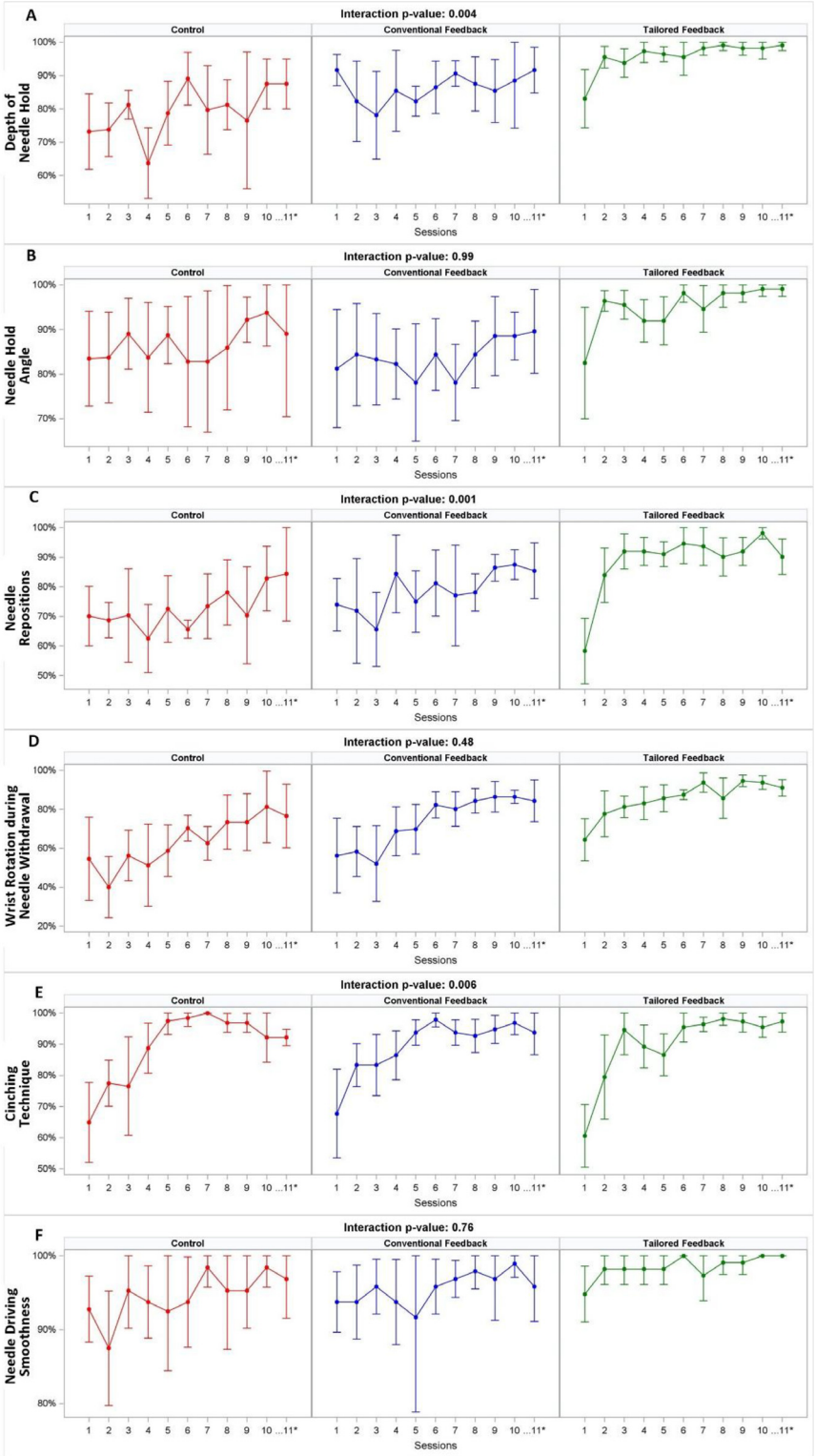


FIGURE 6. Trajectory of top EASE skills through 10 training sessions. The suturing technical skills that best predicted watertightness were compared across the 3 groups based on their rates of ideal EASE scores.

drops in *Depth of Needle Hold*, *Needle Hold Angle*, *Cinching Technique*, and *Needle Driving Smoothness* in the *Tailored* group.

DISCUSSION

Summary

In this study, *Tailored* feedback, defined as feedback on only deficient skills, significantly enhanced VUA watertightness. The targeted technical skills improved in parallel. These findings support teaching complex procedures using personalized, skills-focused approaches.

Result 1: Achieving Watertightness Within 10 Sessions

Our results indicate that all participants in the *Tailored* feedback group achieved and maintained successful watertightness. They improved by session 5, with all reaching watertightness by session 8, sustained through the retention session. Additionally, we identify the most significant suturing skills that could guide future studies focusing on key skills driving watertightness.

Result 2: Top 3 Predictors for Watertight VUA

For the *Tailored* feedback group, 2 of the 3 top predictors for watertightness— *Needle Hold Depth* and *Needle Repositions*—significantly improved more than in the other groups. The third predictor, *Needle Hold Angle*, did not exhibit significant improvement across the full duration of the study, but rapidly improved from sessions 1 to 2, suggesting the interaction test may not capture early improvement. Although the small sample limited statistical power, clear trends indicated that the *Tailored* feedback group still improved.

Result 3: Skill Retention Study

Prior work shows skills decay after laparoscopic suturing training, with more than 50% decay in the first month (De Win,[11]). In our study, we observed no decay across any group approximately 1 month post-training. *Tailored* feedback, participants retained proficiency once acquired, while those who did not reach ideal performance during training did not improve further.

Limitations

While our study demonstrated an overall statistical advantage for the *Tailored* feedback group in watertightness and key suturing skills, it was underpowered to demonstrate differences in some of the more granular suturing skills in our EASE suturing rubric.

Given that our study sample size was small, baseline equivalence randomization was difficult. Baseline suturing performance favored the *Conventional* group. The *Tailored* group demonstrated statistically significant improvements over both the *Conventional* and *Control* groups. This suggests the intervention drove the improvements rather than baseline differences. A larger study is planned to further evaluate these metrics.

Interim practice between sessions was not formally controlled. Opportunities for additional robotic suturing specific to the VUA were limited to a few senior residents, and no participants had access to our simulation outside the study. Although 2 participants dropped out after baseline, their average skill levels suggested attrition was random and unlikely to bias results. The strong statistical significance indicates the study's conclusions remain robust.

Although silicone models provide standardization and have been validated for simulating VUAs in robotic surgical training, their mechanical properties differ from live tissue.[5] The minimal leakage observed in our experiments suggests trainees are capable of watertight closures, but live tissue may behave differently. Our findings demonstrate utility in silicone models for assessing suturing performance, but caution is warranted when comparing to live tissue.

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SUPPLEMENTARY INFORMATION

Supplementary material associated with this article can be found in the online version at doi:10.1016/j.jsurg.2026.103937.